

## PROBLEM SET 4—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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**Instructions:** Due on 11/11/19 before the final starts. This problem set is optional. I will take the average over four problem sets to compute your assignments grade. If you deliver this one it will be extra. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission. Don't forget to attach the cover sheet.

1. You have seen this problem several times in the past problem sets (PS1 and PS3). Let the agent be a drinker who plans to buy at most one bottle of wine within a month. His utility is  $U = \theta q - t$  where  $q$  is the quality he buys and  $\theta$  is a positive parameter that indexes his taste for quality. If he decides not to buy any wine, his utility is just 0. This time, suppose that the taste for quality  $\theta$  is uniformly distributed on the interval  $[\theta_L, \theta_H]$ , where  $0 < \theta_L < \theta_H$

The principal is a local monopoly on the wine market. He can produce wine with any quality  $q \geq 0$ ; the production of a bottle of wine with quality  $q$  costs him  $c(q) = \frac{q^2}{2}$ .

- (a) Characterize the second-best optimal quality-pricing schedule.
- (b) Now suppose that  $\theta$  is distributed on  $[\theta_L, \theta_H]$  according to the distribution  $f(\theta) = \frac{2(\theta_H - \theta)}{(\theta_H - \theta_L)^2}$ . Characterize the second-best optimal quality-pricing schedule in this case. Is the hazard rate function  $\frac{1-F(\theta)}{f(\theta)}$  decreasing? What does a decreasing hazard rate has to do with the monotonicity of the quality-pricing schedule?

### Solution

This is a monopoly screening problem. We covered the general case during my lectures so I will skip several details and refer you to my slides when needed. I will solve the problem first for an arbitrary distribution  $f(\theta)$ . The monopolist problem is

$$\begin{aligned} \max_{(q,t)} \quad & \int_{\theta_L}^{\theta_H} \left( t(\theta) - \frac{q(\theta)^2}{2} \right) f(\theta) d\theta \\ \text{subject to} \quad & \theta q(\theta) - t(\theta) \geq 0 \quad \forall \theta \in [\theta_L, \theta_H] \\ & \theta q(\theta) - t(\theta) \geq \theta q(\theta') - t(\theta') \quad \forall \theta, \theta' \in [\theta_L, \theta_H] \end{aligned}$$

For any contract  $(q, t)$  define  $U(\theta) = \theta q(\theta) - t(\theta)$ , i.e., the utility from the consumer of choosing a contract designed for his true type. A contract  $(q, t)$  is incentive compatible if and only if

- (a)  $q$  is monotone increasing
- (b)  $U(\theta) = U(\theta') + \int_{\theta'}^{\theta} q(\tau) d\tau$ .

You can check my slides for the proof of why the two conditions above are equivalent to incentive compatibility. Note that the above conditions imply that the only participation constraint that matters is the one for the lowest type  $\theta_L$ . Once again, for the proof you can refer to the slides.

The problem of the monopolist is equivalent to

$$\begin{aligned}
& \max_{(q,t)} \int_{\theta_L}^{\theta_H} \left( t(\theta) - \frac{q(\theta)^2}{2} \right) f(\theta) d\theta \\
& \text{subject to } U(\theta) = U(\theta_L) + \int_{\theta_L}^{\theta} q(\tau) d\tau \\
& q \text{ is increasing} \\
& U(\theta_L) \geq 0.
\end{aligned}$$

Note that we can write using the envelope condition the transfers as  $t(\theta) = \theta q(\theta) - U(\theta_L) - \int_{\theta_L}^{\theta} q(\tau) d\tau$ . We can use this fact to rewrite the objective function, which after a lengthy simplification (in the slides, it is the part that involves simplifying the double integral that appears) yields the following problem for the principal

$$\begin{aligned}
& \max_q \int_{\theta_L}^{\theta_H} \left( \theta q(\theta) - \frac{q(\theta)^2}{2} - \frac{1 - F(\theta)}{f(\theta)} q(\theta) \right) f(\theta) d\theta \\
& \text{subject to } q \text{ increasing}
\end{aligned}$$

As usual, we will solve the problem by first ignoring the monotonicity condition and then identifying conditions under which it will hold.

We can maximize the above integral by choosing  $q$  such that it maximizes *pointwise* the integrand, which yields the first order condition:

$$\theta - q(\theta) - \frac{1 - F(\theta)}{f(\theta)} = 0.$$

Solving for  $q$  yields

$$q(\theta) = \max \left\{ \theta - \frac{1 - F(\theta)}{f(\theta)}, 0 \right\},$$

where the max takes care of the possibility of negative solutions. Define  $\theta^*$  as the value that solves the following equation:

$$\theta^* = \frac{1 - F(\theta^*)}{f(\theta^*)}.$$

Then we can express  $q(\theta)$  as

$$q(\theta) = \begin{cases} \theta - \frac{1 - F(\theta)}{f(\theta)} & \theta \in [\theta^*, \theta_H] \\ 0 & \theta \in [\theta_L, \theta^*] \end{cases}$$

Thus, notice that the contract exhibits shutdown for some types.

To calculate the transfers we first compute  $U(\theta)$  :

$$U(\theta) = \int_{\theta_L}^{\theta} \max \left\{ \tau - \frac{1 - F(\tau)}{f(\tau)}, 0 \right\} d\tau = \begin{cases} 0 & \theta \in [\theta_L, \theta^*] \\ \int_{\theta^*}^{\theta} \left( \tau - \frac{1 - F(\tau)}{f(\tau)} \right) d\tau & \theta \in [\theta^*, \theta_H] \end{cases}$$

Thus, transfers are  $t(\theta) = \theta q(\theta) - U(\theta)$  where  $q$  and  $U$  are given by the formulas above.

(a) For the uniform distribution we have that

$$\begin{aligned}
f(\theta) &= \frac{1}{\theta_H - \theta_L} \\
F(\theta) &= \frac{\theta - \theta_L}{\theta_H - \theta_L} \\
\frac{1 - F(\theta)}{f(\theta)} &= \theta_H - \theta.
\end{aligned}$$

Thus  $\theta^* = \frac{\theta_H}{2}$  so that

$$q(\theta) = \begin{cases} 0 & \theta \in [0, \theta_H/2] \\ 2\theta - \theta_H & \theta \in [\theta_H/2, \theta_H] \end{cases}$$

$$t(\theta) = \begin{cases} 0 & \theta \in [0, \theta_H/2] \\ \theta^2 - \frac{\theta_H^2}{4} & \theta \in [\theta_H/2, \theta_H] \end{cases}$$

(b) For this case we have

$$f(\theta) = \frac{2(\theta_H - \theta)}{(\theta_H - \theta_L)^2}$$

$$F(\theta) = \frac{2\theta_H\theta - \theta^2 - 2\theta_H\theta_L + \theta_L^2}{(\theta_H - \theta_L)^2}$$

$$\frac{1 - F(\theta)}{f(\theta)} = \frac{\theta_H - \theta}{2}.$$

Thus  $\theta^* = \frac{\theta_H}{3}$  so that

$$q(\theta) = \begin{cases} 0 & \theta \in [0, \theta_H/3] \\ \frac{3\theta - \theta_H}{2} & \theta \in [\theta_H/3, \theta_H] \end{cases}$$

$$t(\theta) = \begin{cases} 0 & \theta \in [0, \theta_H/3] \\ \frac{9\theta^2 - \theta_H^2}{12} & \theta \in [\theta_H/3, \theta_H] \end{cases}$$

Notice that the hazard rate for this case is decreasing. Recall that a decreasing hazard rate is part of the sufficient conditions required for monotonicity of  $q$ . For the case in this item,  $q$  is indeed increasing in the type.

2. You had a similar problem to this one in the exam. A government agency writes a procurement contract with a firm to deliver  $q$  units of a good. The firm has constant marginal cost  $c$ , so that its profit is  $P - cq$ , where  $P$  denotes the payment for the transaction. The firm's cost is private information and the agency's prior belief about it is uniformly distributed on the interval  $[\underline{c}, \bar{c}]$ . The agency makes a take-it-or-leave-it offer to the firm (whose default profit is zero). Suppose that  $B(q)$  denotes the benefit to the agency of obtaining  $q$  units.  $B(q)$  is a strictly increasing and a strictly concave function of  $q$ .

- (a) Characterize the first-best contract.  
(b) Characterize the second-best contract.

#### Solution

- (a) Let  $(q(c), P(c))$  be a contract. At the first-best, the principal solves, for any  $c \in [\underline{c}, \bar{c}]$

$$\max_{q(c), P(c)} B(q(c)) - P(c)$$

subject to  $P(c) - cq(c) \geq 0$ .

At an optimal solution, the participation constraint must be binding, i.e.,  $P(c) = cq(c)$ . Substituting the binding constraint into the objective yields, for any  $c \in [\underline{c}, \bar{c}]$ , the following problem

$$\max_{q(c)} B(q(c)) - cq(c)$$

The first order condition for the above problem is

$$B'(q(c)) = c \quad \forall c \in [\underline{c}, \bar{c}]$$

which characterizes the first-best solution for the principal.

- (b) I will use here a general distribution function  $f(c)$  and will specialize to the case of the uniform later. The optimal second-best contract is given by the solution to

$$\begin{aligned} \max_{q(c), P(c)} \quad & \int_{\underline{c}}^{\bar{c}} (B(q(c)) - P(c)) f(c) dc \\ & P(c) - cq(c) \geq 0, \forall c \\ & P(c) - cq(c) \geq P(c') - cq(c'), \forall c, c' \end{aligned}$$

Define  $U(c) = P(c) - cq(c)$ . I will first show that incentive compatibility constraints are satisfied if and only if

- i.  $q$  is decreasing
- ii.  $U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau$  for any  $c$

*Proof.* Using the definition of  $U(c)$  we can write the incentive compatibility constraints for any  $c' > c$  as follows:

$$\begin{aligned} U(c) &\geq U(c') + q(c')(c' - c) \\ U(c') &\geq U(c) + q(c)(c - c') \end{aligned}$$

Equivalently,

$$q(c')(c' - c) \leq U(c) - U(c') \leq q(c)(c' - c)$$

Therefore,  $c' > c$  implies that  $q(c) \geq q(c')$ . Dividing by  $c' - c$  and taking the limit as  $c' \rightarrow c$ , we get

$$\frac{dU(c)}{dc} = -q(c)$$

Therefore

$$U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau$$

Suppose now that  $q$  is decreasing and  $U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau$ . Let  $c' > c$ . we have

$$\begin{aligned} U(c) - U(c') &= \int_{\underline{c}}^{c'} q(\tau) d\tau - \int_{\underline{c}}^c q(\tau) d\tau \\ &= \int_c^{c'} q(\tau) d\tau \\ &\geq q(c')(c' - c) \end{aligned}$$

and hence the IC constraints are satisfied. □

Notice that the participation constraint of type  $\bar{c}$ , i.e.,  $U(\bar{c}) \geq 0$  implies that

$$U(\underline{c}) - \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau = U(\bar{c}) \geq 0.$$

Therefore,

$$U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau \geq \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau - \int_{\underline{c}}^c q(\tau) d\tau = \int_c^{\bar{c}} q(\tau) d\tau \geq 0,$$

Where the first inequality is by the participation constraint of type  $\bar{c}$  and the second inequality is by the fact that we are integrating a positive function. Thus, in other words, the participation constraint of type  $\bar{c}$  implies all the other participation constraints.

Therefore, the problem is equivalent to

$$\begin{aligned} & \max_{q(c), P(c)} \int_{\underline{c}}^{\bar{c}} (B(q(c)) - P(c)) f(c) dc \\ & \text{subject to } U(\bar{c}) \geq 0 \\ & U(c) = U(\underline{c}) - \int_{\underline{c}}^c q(\tau) d\tau \\ & q \text{ is decreasing} \end{aligned}$$

Using the definition of  $U(c)$  and substituting we can write the objective function as

$$\int_{\underline{c}}^{\bar{c}} \left[ B(q(c)) - cq(c) - U(\underline{c}) + \int_{\underline{c}}^c q(\tau) d\tau \right] f(c) dc$$

Substituting for  $U(\underline{c})$  using

$$U(\underline{c}) = U(\bar{c}) + \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau,$$

we can write the objective function as

$$\begin{aligned} & \int_{\underline{c}}^{\bar{c}} \left[ B(q(c)) - cq(c) - U(\bar{c}) - \int_{\underline{c}}^{\bar{c}} q(\tau) d\tau + \int_{\underline{c}}^c q(\tau) d\tau \right] f(c) dc \\ & = \int_{\underline{c}}^{\bar{c}} \left[ B(q(c)) - cq(c) - \int_c^{\bar{c}} q(\tau) d\tau \right] f(c) dc - U(\bar{c}) \end{aligned}$$

At a solution, we must have that  $U(\bar{c}) = 0$ . We need now to simplify the double integral

$$\int_{\underline{c}}^{\bar{c}} \left( \int_c^{\bar{c}} q(\tau) d\tau \right) f(c) dc.$$

Using integration by parts this is equal to

$$\int_{\underline{c}}^{\bar{c}} F(c) q(c) dc.$$

The problem is therefore

$$\int_{\underline{c}}^{\bar{c}} \left[ B(q(c)) - cq(c) - \frac{F(c)}{f(c)} q(c) \right] f(c) dc$$

subject to the monotonicity constraint. As usual, we will ignore the monotonicity constraint and then show that the solution satisfies it. The first order condition for each  $c$  is given by

$$B'(q(c)) = c + \frac{F(c)}{f(c)}$$

Notice that at the lowest cost  $\underline{c}$ , the solution is identical to the first-best as was expected. Now, for the uniform distribution,  $F(c) = \frac{c-\underline{c}}{\bar{c}-\underline{c}}$  and  $F(c)/f(c) = c - \underline{c}$  so that

$$B'(q(c)) = c + c - \underline{c} = 2c - \underline{c}$$

Therefore

$$q(c) = (B')^{-1} (2c - \underline{c})$$

Now, if  $B'(q) = b$  then  $((B')^{-1})'(b) = \frac{1}{B''(q)} = \frac{1}{B''((B')^{-1}(b))} < 0$  for any  $b$  since  $B'' < 0$ . Thus, we have that

$$\frac{dq(c)}{dc} = ((B')^{-1})'(2c - \underline{c}) (2) < 0$$

and therefore,  $q$  is strictly decreasing.

Notice also that since  $B' > 0$ , we have that  $q > 0$  for every  $c$ . Finally, note that we can compute the transfers by first calculating  $U(c)$  and then using  $P(c) = U(c) + cq(c)$ .

3. Consider a principal who hires a salesman to approach a client. The reservation utility level of the salesman is 3. A high level of effort gives no sale with probability 0.2, a \$100 sale with probability 0.4 and a \$400 sale with probability 0.4. For a low level of effort, the three probabilities are 0.4, 0.4, 0.2 respectively.

The principal is risk neutral. The agent is risk averse, and if the wage is  $w$  and the effort is  $a$ , he gets utility  $u(w, a) = \sqrt{w} - a$ . (High effort requires  $a = 3$  and low effort requires  $a = 0$ .)

- For starters, assume that the principal could observe the level of effort put by the salesman. Characterize the first-best wage schedule.
- From now on, assume that contracts can be made contingent on the size of the sale only (effort is unobservable). What is the optimal way to induce the salesman to put in a low level of effort?
- What is the optimal way to induce the salesman to put in a high level of effort?
- What is the optimal contract (from the principal's perspective) to offer the salesman?

### Solution

Note that in this problem we will have an *implicit limited liability constraint* because of the domain of the square root is the positive real numbers. Thus any wage schedule needs to be positive. This will not cause problems until part (c), so we will proceed without the non-negativity constraint on wages in all of the problems until (c).

- (a) A principal that would like to induce a low level of effort at the first-best solves

$$\begin{aligned} \max_{(w_l, w_m, w_h) \in \mathbb{R}_+^3} \quad & 0.4(0 - w_l) + 0.4(100 - w_m) + 0.2(400 - w_h) \\ \text{subject to} \quad & 0.4\sqrt{w_l} + 0.4\sqrt{w_m} + 0.2\sqrt{w_h} \geq 3 \end{aligned}$$

The Lagrangian of this problem is

$$\mathcal{L} = 0.4(0 - w_l) + 0.4(100 - w_m) + 0.2(400 - w_h) + \lambda(0.4\sqrt{w_l} + 0.4\sqrt{w_m} + 0.2\sqrt{w_h} - 3)$$

The Kuhn-Tucker conditions are

$$\begin{aligned} w_l : \quad & -0.4 + \frac{0.4\lambda}{2\sqrt{w_l}} = 0 \\ w_m : \quad & -0.4 + \frac{0.4\lambda}{2\sqrt{w_m}} = 0 \\ w_h : \quad & -0.2 + \frac{0.2\lambda}{2\sqrt{w_h}} = 0 \\ \lambda : \quad & 0.4\sqrt{w_l} + 0.4\sqrt{w_m} + 0.2\sqrt{w_h} \geq 3, \lambda \geq 0, \lambda[0.4\sqrt{w_l} + 0.4\sqrt{w_m} + 0.2\sqrt{w_h} - 3] = 0 \end{aligned}$$

From the first constraint we must have that  $\lambda > 0$ , i.e., the PC constraint is binding. From the first three constraints we have that

$$\lambda = \frac{1}{2\sqrt{w_i}}, \quad \text{for } i = l, m, h$$

Monotonicity of the square root function then implies that  $w_l = w_m = w_h$ ; denote this common wage as  $\underline{w}$ . From the binding participation constraint we have that  $\sqrt{\underline{w}} = 3$  which implies that  $\underline{w} = 9$ . Optimal profits are \$111.

On the other hand, a principal that would like to induce a high level of effort at the first-best solves

$$\begin{aligned} \max_{(w_l, w_m, w_h) \in \mathbb{R}_+^3} \quad & 0.2(0 - w_l) + 0.4(100 - w_m) + 0.4(400 - w_h) \\ \text{subject to} \quad & 0.2\sqrt{w_l} + 0.4\sqrt{w_m} + 0.4\sqrt{w_h} - 3 \geq 3 \end{aligned}$$

The solution for this problem is similar to the previous one, there will be a constant wage and the participation constraint must be binding. Denote as  $\bar{w}$  as the constant wage solution to this problem. Then from the binding participation constraint we have that  $\sqrt{\bar{w}} = 6$  so that  $\bar{w} = 36$ . Optimal profits are \$164.

Therefore, the principal prefers to induce a high-level of effort at the first-best.

(b) The problem of the principal is

$$\begin{aligned} \max_{(w_l, w_m, w_h) \in \mathbb{R}_+^3} \quad & 0.4(0 - w_l) + 0.4(100 - w_m) + 0.2(400 - w_h) \\ \text{subject to} \quad & 0.4\sqrt{w_l} + 0.4\sqrt{w_m} + 0.2\sqrt{w_h} \geq 3 \\ & 0.4\sqrt{w_l} + 0.4\sqrt{w_m} + 0.2\sqrt{w_h} \geq 0.2\sqrt{w_l} + 0.4\sqrt{w_m} + 0.4\sqrt{w_h} - 3 \end{aligned}$$

The Lagrangian is

$$\begin{aligned} \mathcal{L} = & -0.4(w_l) + 0.4(100 - w_m) + 0.2(400 - w_h) + \mu[0.4\sqrt{w_l} + 0.4\sqrt{w_m} + 0.2\sqrt{w_h} - 3] \\ & + \lambda[0.2(\sqrt{w_l} - \sqrt{w_h}) + 3] \end{aligned}$$

The Kuhn-Tucker conditions are

$$\begin{aligned} w_l : \quad & -0.4 + \frac{0.4\mu}{2\sqrt{w_l}} + \frac{0.2\lambda}{2\sqrt{w_l}} = 0 \Leftrightarrow \mu + 0.5\lambda = 2\sqrt{w_l} \\ w_m : \quad & -0.4 + \frac{0.4\mu}{2\sqrt{w_m}} = 0 \Leftrightarrow \mu = 2\sqrt{w_m} \\ w_h : \quad & -0.2 + \frac{0.2\mu}{2\sqrt{w_h}} - \frac{0.2\lambda}{2\sqrt{w_h}} = 0 \Leftrightarrow \mu - \lambda = 2\sqrt{w_h} \\ \mu : \quad & 0.4\sqrt{w_l} + 0.4\sqrt{w_m} + 0.2\sqrt{w_h} \geq 3, \mu \geq 0, \mu[0.4\sqrt{w_l} + 0.4\sqrt{w_m} + 0.4\sqrt{w_h} - 3] = 0 \\ \lambda : \quad & 0.2(\sqrt{w_l} - \sqrt{w_h}) + 3 \geq 0, \lambda \geq 0, \lambda[0.2(\sqrt{w_l} - \sqrt{w_h}) + 3] = 0 \end{aligned}$$

From the second condition with respect to  $w_m$  we see that we must have  $\mu > 0$ . Thus, the participation constraint is binding. Now, towards a contradiction, suppose that  $\lambda > 0$ . Then from the condition on  $\lambda$  we must have that  $\sqrt{w_h} - \sqrt{w_l} = 15$ . From the first and the third conditions then we have that  $\sqrt{w_h} - \sqrt{w_l} = -(3/4)\lambda < 0$  which is a contradiction to the previous statement. Thus,  $\lambda = 0$  which implies the IC constraint is not binding. Since  $\lambda = 0$ , this implies that the solution is the same as the first-best solution when the principal wants to induce a low-level of effort that we identified before. Thus, the wages should be constant and equal to  $\underline{w} = 9$ . Profits are \$111.

(c) For this problem I will add non-negativity constraints (what I mentioned was akin to a limited liability constraint). If you try to solve the problem without them, you will surely run into trouble! Also, the analysis will be much simpler if we do the transformation to the variables of the problem that I suggested in my lecture slides. Thus, define for each  $i = l, m, h$ , the utility of wage  $w_i$  as  $x_i = \sqrt{w_i}$  so that  $x_i^2 = w_i$ . The principal's problem is:

$$\begin{aligned} \max_{(x_l, x_m, x_h) \in \mathbb{R}^3} \quad & 0.2(0 - x_l^2) + 0.4(100 - x_m^2) + 0.4(400 - x_h^2) \\ \text{subject to} \quad & 0.2x_l + 0.4x_m + 0.4x_h - 3 \geq 3 \\ & 0.2x_l + 0.4x_m + 0.4x_h - 3 \geq 0.2x_l + 0.4x_m + 0.2x_h \\ & x_l \geq 0, x_m \geq 0, x_h \geq 0 \end{aligned}$$

The Lagrangian is

$$\begin{aligned}\mathcal{L} = & -0.2x_l^2 + 0.4(100 - x_m^2) + 0.4(400 - x_h^2) + \mu[0.2x_l + 0.4x_m + 0.4x_h - 6] \\ & + \lambda[0.2(x_h - x_l) - 3] + \xi_l x_l + \xi_m x_m + \xi_h x_h\end{aligned}$$

The Kuhn-Tucker conditions are

$$x_l : -0.4x_l + 0.2\mu - 0.2\lambda + \xi_l = 0 \quad (1)$$

$$x_m : -0.8x_m + 0.4\mu + \xi_m = 0 \quad (2)$$

$$x_h : -0.8x_h + 0.4\mu + 0.2\lambda + \xi_h = 0 \quad (3)$$

$$\mu : \mu \geq 0, \quad 0.2x_l + 0.4x_m + 0.4x_h \geq 6, \quad \mu[0.2x_l + 0.4x_m + 0.4x_h - 6] = 0 \quad (4)$$

$$\lambda : \lambda \geq 0, \quad 0.2(x_h - x_l) \geq 3, \quad \lambda[0.2(x_h - x_l) - 3] = 0 \quad (5)$$

$$\xi_l : x_l \geq 0, \xi_l \geq 0, \xi_l x_l = 0 \quad (6)$$

$$\xi_m : x_m \geq 0, \xi_m \geq 0, \xi_m x_m = 0 \quad (7)$$

$$\xi_h : x_h \geq 0, \xi_h \geq 0, \xi_h x_h = 0 \quad (8)$$

We have several cases to consider, which depend on the sign of the variables  $x_i$ :

- i.  $x_l = x_m = x_h = 0$ . Notice that this case violates the IC constraint (5) so this cannot be a solution to the KT conditions.
- ii.  $x_l > 0, x_m > 0, x_h > 0$ . This implies from (6), (7) and (8) that  $\xi_l = \xi_m = \xi_h = 0$ . Thus, we have that

$$0.4x_l = 0.2\mu - 0.2\lambda \quad (9)$$

$$0.8x_m = 0.4\mu \Leftrightarrow \mu = 2x_m \Rightarrow \mu > 0. \quad (10)$$

$$0.4x_h = 0.4\mu + 0.2\lambda \quad (11)$$

Since  $\mu > 0$ , this implies that the participation constraint is binding, i.e.

$$0.2x_l + 0.4x_m + 0.4x_h = 6. \quad (12)$$

Plugging (10) into (9) and (11) yields

$$x_l = x_m - 0.5\lambda \quad (13)$$

$$x_h = x_m + 0.5\lambda \quad (14)$$

Plugging (13) and (14) into (12) yields that  $x_m = 6$  which in turn implies from (10) that  $\mu = 12$ . The fact that  $x_m = 6$  implies that

$$x_l = 6 - 0.5\lambda \quad (15)$$

$$x_h = 6 + 0.5\lambda \quad (16)$$

$$0.2x_l + 0.4x_h = 3.6 \quad \text{from (12)} \quad (17)$$

Recall that from condition (5),  $\lambda \geq 0$ . From the previous two equations we need to analyze to sub-cases. If  $\lambda = 0$  then  $x_l = x_h = 6$  but this implies from the incentive compatibility constraint that  $0 \geq 3$  which is absurd. On the other hand, if  $\lambda > 0$  this implies that the IC constraint is binding so that

$$x_h = 15 + x_l. \quad (18)$$

Equations (17) and (18) together imply that  $x_l = -4$  and  $x_h = 11$  which of course is also absurd since we started under the assumption that all variables were positive. Thus, this whole case yields a contradiction so this cannot be a solution to the KT conditions.

- iii.  $x_l = 0, x_m > 0, x_h > 0$ . This implies that  $\xi_m = \xi_h = 0$ . This implies from (2) that  $\mu > 0$  so that the participation constraint is binding, i.e.,

$$0.4x_m + 0.4x_h = 6 \quad (19)$$

Also, from (1) we get that

$$0.2\lambda = 0.2\mu + \xi_l \quad (20)$$

which implies that  $\lambda > 0$  since  $\mu > 0$ . This in turn implies that the IC constraint is binding which implies that  $x_h = 15$  since  $x_l = 0$ . But with these values, the binding participation constraint (19) implies that  $x_m = 0$  which is a contradiction from our initial assumption. Thus, this also cannot be a solution to the KT conditions.

- iv.  $x_l > 0, x_m = 0, x_h > 0$ . This implies that  $\xi_l = \xi_h = 0$ . From (3) we get that

$$0.4\mu + \xi_m = 0 \quad (21)$$

Since the multipliers need to be positive, equation (21) implies that  $\mu = \xi_m = 0$ . From equations (1) and (3) this implies that  $x_l = -0.5\lambda$  and  $x_h = 0.25\lambda$ . Since both variables need to be positive the previous equations imply that  $\lambda = 0$  but this implies that  $x_l = x_h = 0$  which again contradicts the initial assumption. Thus, this also cannot be a solution to the KT conditions.

- v.  $x_l > 0, x_m > 0, x_h = 0$ . This implies that  $\xi_l = \xi_m = 0$ . From (3) we get that  $0.4\mu + 0.2\lambda + \xi_h = 0$  which implies that  $\mu = \lambda = \xi_h = 0$  since all multipliers need to be positive. However, (1) and (2) then imply  $x_l = x_m = 0$  which again is a contradiction, so this cannot be a solution to the KT conditions.
- vi.  $x_l > 0, x_m = x_h = 0$ . This implies that  $\xi_l = 0$ . Equation (3) once again implies that  $\mu = \xi_m = \lambda = 0$  which in turn implies, from equation (1) that  $x_l = 0$  and once again we reach a contradiction. Thus, this cannot be a solution to the KT conditions.
- vii.  $x_l = x_h = 0, x_m > 0$ . This implies that  $\xi = 0$ . Equation (2) implies that  $\mu > 0$ . Once again, (3) implies that  $\mu = \xi_m = \lambda = 0$  which contradicts the previous statement. Thus this cannot be a solution to the KT conditions.
- viii.  $x_l = x_m = 0, x_h > 0$ . This is the last case, so as you might have already guessed, this will be the only solution to the KT conditions. Let us prove this.  $x_h > 0$  implies that  $\xi_h = 0$ . Equation (2) implies that  $\mu = \xi_m = 0$ . Equation (3) implies that  $x_h = 0.25\lambda$  and since  $x_h > 0$  this implies that  $\lambda > 0$  and that the IC constraint is binding. The binding IC constraint implies that  $x_h = 15$  and that  $\lambda = 60$  from the previous expression. Finally from (1) we get, by plugging all the values, that  $\xi_l = 12$ . All these values satisfy the KT conditions so this is a solution to the problem.

As we found out, the solution is  $x_l = x_m = 0, x_h = 15$  so that  $w_l = w_m = 0$  and  $w_h = 225$ . Profits from this wage schedule are \$110.

- (d) The previous calculations show that the maximum profit is obtained when the principal induces a *low level of effort* with a constant wage schedule of  $\underline{w} = 9$  which yields a profit of \$111.

4. Consider the following moral hazard insurance model. Let the consumer's utility function for wealth be  $u(w) = \sqrt{w}$ , his initial wealth be  $w_0 = \$100$ , and suppose that there are two loss levels,  $l = 0$  and  $l = \$51$ . As usual, there are two effort levels  $e = 0$  and  $e = 1$ . The consumer's disutility of effort is given by the function  $d(e)$ , where  $d(0) = 0$  and  $d(1) = 1/3$ . Finally, suppose that the loss probabilities are given by the following table, in which rows correspond to effort and column to loss levels:

|         | $l = 0$ | $l = 51$ |
|---------|---------|----------|
| $e = 0$ | 1/3     | 2/3      |
| $e = 1$ | 2/3     | 1/3      |

- (a) Verify that the probabilities given in the table satisfy the monotone likelihood ratio property.
- (b) Find the consumer's reservation utility assuming that there is only one insurance company and that the consumer's only other option is to self-insure. *Hint: this exercise provides a justification of where the reservation utility comes from. You need to compute what level of utility the consumer can achieve without insurance.*
- (c) What effort level will the consumer exert if no insurance is available? *Hint: This continues the previous part. Without insurance, the consumer will exert the effort level that achieves his reservation utility.*
- (d) Show that if information is symmetric (first-best), then it is optimal for the insurance company to offer a policy that induces high effort.
- (e) Show that the policy in part (d) will not induce high effort if information is asymmetric.

- (f) Find the optimal policy when information is asymmetric.
- (g) Compare the insurance company's profits in the symmetric and asymmetric information cases. Also, compare the consumer's utility in the two cases. Argue that the symmetric information solution Pareto dominates (is more efficient) that with asymmetric information.

### Solution

- (a) The monotone likelihood ration property says that the ratio  $\frac{\Pr(l=1|e=1)}{\Pr(l=1|e=0)}$  should be decreasing on the level of the loss, i.e., higher effort make higher losses less likely. The likelihood ratios are

$$\begin{aligned}\frac{\Pr(l=0|e=1)}{\Pr(l=0|e=0)} &= \frac{2/3}{1/3} = 2 \\ \frac{\Pr(l=51|e=1)}{\Pr(l=51|e=0)} &= \frac{1/3}{2/3} = \frac{1}{2}\end{aligned}$$

As we can see the likelihood ratio is indeed decreasing with respect to an increasing loss.

- (b) If the consumer can't insure, we need to compute his expected utility in the case of low and high level of effort:

$$\text{For low effort: } (1/3)\sqrt{100-0} + (2/3)\sqrt{100-51} = 8$$

$$\text{For high effort: } (2/3)\sqrt{100-0} + (1/3)\sqrt{100-51} - 1/3 = \frac{26}{3} \approx 8.67$$

Since the expected utility with high effort is larger than the one with low effort, it is optimal for the consumer to choose a high level of effort. Thus, his reservation utility  $\underline{U} = 26/3$ .

- (c) Following on the previous item calculation, if no insurance is available, the agent will induce a high level of effort and get a utility of 8.67.
- (d) An insurance contract specifies a premium  $P$  and the amount of benefits  $B$  to be reimbursed in the case of a loss. Thus, a insurer that wants to induce a high level of effort under the first-best would solve the problem

$$\begin{aligned}\max_{P,B} \quad & P - (1/3)B \\ \text{subject to} \quad & (2/3)\sqrt{100-P} + (1/3)\sqrt{49-P+B} - 1/3 \geq 26/3.\end{aligned}$$

Notice that because of risk aversion from the consumer, for a fixed premium  $P$  we have that

$$\sqrt{100-P} > (2/3)\sqrt{100-P} + (1/3)\sqrt{49-P+B} \quad \forall B \in [0, 51],$$

that is, a level of reimbursement of  $B = 51$ , which equals the total amount of the loss, always gives a strictly higher expected utility with respect to any other level of reimbursement less than 51. Therefore, at an optimum, the insurer will cover all damages but then will extract the whole surplus of the consumer by charging a high enough premium. Therefore, the participation constraint must hold with equality and the optimal premium level solves  $\sqrt{100-P} - 1/3 = 26/3 \Rightarrow P = 19$ . Thus, the optimal first-best insurance contract that induces the agent to put effort is  $(P, B) = (19, 51)$ . Profits from this insurance contract are \$2.

On the other hand, an insurer that wants to induce a low-level of effort under the first-best would solve

$$\begin{aligned}\max_{P,B} \quad & P - (2/3)B \\ \text{subject to} \quad & (1/3)\sqrt{100-P} + (2/3)\sqrt{49-P+B} \geq 26/3.\end{aligned}$$

Once again, the optimal reimbursement needs to be  $B = 51$  for the reasons explained above. The optimal premium solves  $\sqrt{100-P} = 26/3 \Rightarrow P = 224/9 \approx 24.89$ . The optimal first-best contract that induces a low level of effort is  $(P, B) = (24.89, 51)$ . Profits from this insurance contract are  $-\$82/9 \approx -\$9.1$ , which is a loss to the insurer.

The previous calculations show that an insurer with symmetric (first-best) information would offer a policy that induces high effort.

(e) Consider the contract  $(P, B) = (18.94, 51)$ . The expected utility from this contract is

$$(2/3)\sqrt{100 - 18.94} + (1/3)\sqrt{49 - 18.94 + 51} - 1/3 = 8.67.$$

However, with the same contract, if the consumer doesn't put effort, his expected utility becomes

$$(1/3)\sqrt{100 - 18.94} + (2/3)\sqrt{49 - 18.94 + 51} = 9$$

Thus with the first-best policy the consumer will prefer to not put effort it cannot be monitored.

(f) Suppose first that the principal wants to induce a high level of effort. Since the insurance contract is restricted from below ( $P$  and  $B$  need to be positive) and from above (in no case the consumer can attain a negative wealth level), it will be convenient to rewrite the variables in the problem. Define  $u_g = \sqrt{100 - P}$  and  $u_b = \sqrt{49 - P + B}$ . Thus we have that  $P = 100 - u_g^2$  and  $B = 51 + u_b^2 - u_g^2$ . The insurer's problem is

$$\begin{aligned} \max_{(u_g, u_b) \in \mathbb{R}_+^2} \quad & 100 - u_g^2 - (1/3)(51 + u_b^2 - u_g^2) \\ \text{subject to} \quad & (2/3)u_g + (1/3)u_b - 1/3 \geq 26/3. \\ & (2/3)u_g + (1/3)u_b - 1/3 \geq (1/3)u_g + (2/3)u_b \end{aligned}$$

The Lagrangian is

$$\mathcal{L} = 100 - u_g^2 - (1/3)(51 + u_b^2 - u_g^2) + \mu[(2/3)u_g + (1/3)u_b - 9] + \lambda[(1/3)(u_g - u_b) - 1/3] + \xi_g u_g + \xi_b u_b$$

The Kuhn-Tucker conditions are

$$u_g : - (4/3)u_g + (2/3)\mu + (1/3)\lambda + \xi_g = 0 \quad (22)$$

$$u_b : - (2/3)u_b + (1/3)\mu - (1/3)\lambda + \xi_b = 0 \quad (23)$$

$$\mu : \mu \geq 0, \quad (2/3)u_g + (1/3)u_b - 9 \geq 0, \quad \mu[(2/3)u_g + (1/3)u_b - 9] = 0 \quad (24)$$

$$\lambda : \lambda \geq 0, \quad (1/3)(u_g - u_b) - 1/3 \geq 0, \quad \lambda[(1/3)(u_g - u_b) - 1/3] = 0 \quad (25)$$

$$\xi_g : u_g \geq 0, \xi_g \geq 0, \xi_g u_g = 0 \quad (26)$$

$$\xi_b : u_b \geq 0, \xi_b \geq 0, \xi_b u_b = 0 \quad (27)$$

First notice that from (25) we have that we must have  $u_g \neq u_b$ . Furthermore, the condition implies that  $u_g > u_b$ . Therefore, we can immediately rule out the cases  $u_b = u_g = 0$  and  $u_b > 0, u_g = 0$ . Consider then the following two cases:

- i.  $u_g > 0$  and  $u_b = 0$ . This implies that  $\xi_g = 0$ . Now, suppose that  $\lambda = 0$ , from (23) this implies that  $\mu = \xi_b = 0$  but this would in turn imply that  $u_g = 0$  which is absurd. On the other hand, suppose that  $\lambda > 0$ , which implies that the IC constraint is binding and thus we obtain that  $u_g = 1$  but this is incompatible with the participation constraint. Therefore, in either case we reach a contradiction and this case cannot constitute a solution to the KT conditions.
- ii.  $u_g > 0$  and  $u_b > 0$ . This implies that  $\xi_g = \xi_b = 0$ . From (23) note that we cannot have  $\mu = 0$  and thus  $\mu > 0$  and the participation constraint is binding. If  $\lambda = 0$  this implies, from (22) and (23), that  $u_g = u_b$  which, as discussed before, would violate the incentive compatibility constraint. Then we must have  $\lambda > 0$  and the incentive compatibility constraint is also binding. The system of equations of these two binding constraints is

$$2u_g + u_b = 27$$

$$u_g - u_b = 1$$

Solving the aforementioned system yields  $u_g = 28/3$  and  $u_b = 25/3$ . The previous expressions imply a premium of  $P = 116/9$  and a reimbursement benefit of  $B = 100/3$ . This insurance contract then yields a profit of  $\$16/9 = \$1.78$ .

Now let us consider an insurer that wants to induce a low-level of effort. The problem is

$$\begin{aligned} \max_{(u_g, u_b) \in \mathbb{R}_+^2} \quad & 100 - u_g^2 - (2/3)(51 + u_b^2 - u_g^2) \\ \text{subject to} \quad & (1/3)u_g + (2/3)u_b \geq 26/3. \\ & (1/3)u_g + (2/3)u_b \geq (2/3)u_g + (1/3)u_b - 1/3 \end{aligned}$$

The Lagrangian is

$$\mathcal{L} = 100 - u_g^2 - (2/3)(51 + u_b^2 - u_g^2) + \mu[(1/3)u_g + (2/3)u_b - 26/3] + \lambda[(1/3)(u_b - u_g) + 1/3] + \xi_g u_g + \xi_b u_b$$

The Kuhn-Tucker conditions are

$$u_g : - (2/3)u_g + (1/3)\mu - (1/3)\lambda + \xi_g = 0 \quad (28)$$

$$u_b : - (4/3)u_b + (2/3)\mu + (1/3)\lambda + \xi_b = 0 \quad (29)$$

$$\mu : \mu \geq 0, \quad (1/3)u_g + (2/3)u_b - 26/3 \geq 0, \quad \mu[(1/3)u_g + (2/3)u_b - 26/3] = 0 \quad (30)$$

$$\lambda : \lambda \geq 0, \quad (1/3)(u_b - u_g) + 1/3 \geq 0, \quad \lambda[(1/3)(u_b - u_g) + 1/3] = 0 \quad (31)$$

$$\xi_g : u_g \geq 0, \xi_g \geq 0, \xi_g u_g = 0 \quad (32)$$

$$\xi_b : u_b \geq 0, \xi_b \geq 0, \xi_b u_b = 0 \quad (33)$$

Once again we have four cases to consider. We can immediately notice that the case  $u_g = u_b = 0$  violates the participation constraint. The remaining cases are

- i.  $u_g = 0, u_b > 0$ . This case implies that  $\xi_b = 0$ . Suppose  $\lambda > 0$ , then the IC constraint is binding, but this implies that  $u_b = 0$  which contradicts our initial assumption. On the other hand if  $\lambda = 0$ , then from (28) we see that this implies that  $\mu = \xi_g = 0$  which in turn implies, using (29), that  $u_b = 0$  which again is inconsistent with our initial assumption. Thus this case does not satisfy the KT conditions.
- ii.  $u_g > 0, u_b = 0$ . This case implies that  $\xi_g = 0$ . From (29) this implies that  $\mu = \lambda = \xi_b = 0$ . But this in turn implies that  $u_g = 0$  which contradicts our initial assumption. Thus this case cannot satisfy the KT conditions.
- iii.  $u_g > 0, u_b > 0$ . This implies that  $\xi_g = \xi_b = 0$ . From (29) we see that this case implies that  $\mu > 0$ . Now we will consider two sub-cases

A.  $\lambda = 0$ , from (28) and (29) that  $u_g = u_b$ . With the binding participation constraint then we would obtain a solution similar to the complete information case, with  $u_g = u_b = 26/3$  and  $P = 24.89, B = 51$  and a loss for the insurer of \$ - 9.1. This case satisfies the KT condition but we will need to compare with the next sub-case to see which one is optimal.

B.  $\lambda > 0$ , then both IC and PC constraints are binding, yielding a system of equations:

$$u_g + 2u_b = 26$$

$$-u_g + u_b = -1.$$

Solving this system yields  $u_g = 28/3, u_b = 25/3, P = 12.89, B = 33.33$  Profits are \$ - 9.33

Both of the sub-cases solve the KT conditions, but only the first one, in which  $P = 24.89, B = 51$  is optimal, since it provides a lower loss.

In any case, we conclude that the optimal contract induces a high-level of effort and has  $P = 116/9$  and a reimbursement benefit of  $B = 100/3$ . This insurance contract then yields a profit of  $\$16/9 = \$1.78$ .

(g) Consider the following table

| Contract ( $P, B$ )           | Profits | Utility |
|-------------------------------|---------|---------|
| (19, 51): (first-best)        | \$2     | 26/3    |
| (116/9, 100/3): (second-best) | \$1.78  | 26/3    |

Notice that the consumer is equal under both the first-best and the second-best. However, the principal is better under the first-best. There is a cost to the principal of inducing a high-level of effort when this can't be monitored. Since the consumer is the same but the principal is better in the first-best contract, then this one Pareto-dominates the second best.